Neuro-Retail

A Customer Behavior Analyzer

End-to-end retail analytics pipeline
From raw data to actionable business insights

GDSC · Machine Learning Track - ML 3

Meet the Team



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01 — INTRODUCTION

Project Overview

Most businesses lose customers simply because they react too late. The Customer Behavior Analyzer is a full-stack ML and Deep Learning platform that goes beyond historical reporting — it actively predicts when each customer will buy next, recommends the right products, detects churn before it happens, and delivers personalized offers at exactly the right moment.

02 — ARCHITECTURE

Tech Stack & Pipeline

PIPELINE FLOW

Raw Data — Cleaning — EDA — Feature Eng. — K-Means — XGBoost/DNN — Insights

TECHNOLOGIES

Python	Pandas	NumPy
Scikit-learn	K-Means	StandardScaler
XGBoost	TensorFlow / Keras	mlxtend (Apriori)
Matplotlib	React+ flask API	Plotly

03 — EDA

Exploratory Data Analysis

1.07M
ORIGINAL ROWS
raw transactions

779K
AFTER CLEANING
288K rows removed

5,878
UNIQUE CUSTOMERS
4,631 products

KEY FINDINGS

- A small group of customers brings in most of the revenue.
- Revenue peaks in Oct-Nov (Black Friday / holiday effect)
- Weekday purchasing far higher than weekends

RFM Analysis

FEATURES CREATED

- TotalPrice = Quantity × Price
- Average order value · purchase frequency
- Inter-purchase time intervals
- Unique products per customer
- Time features: year, month, day, hour

RFM MODEL

R — Recency
Days since last purchase

F — Frequency
Number of transactions

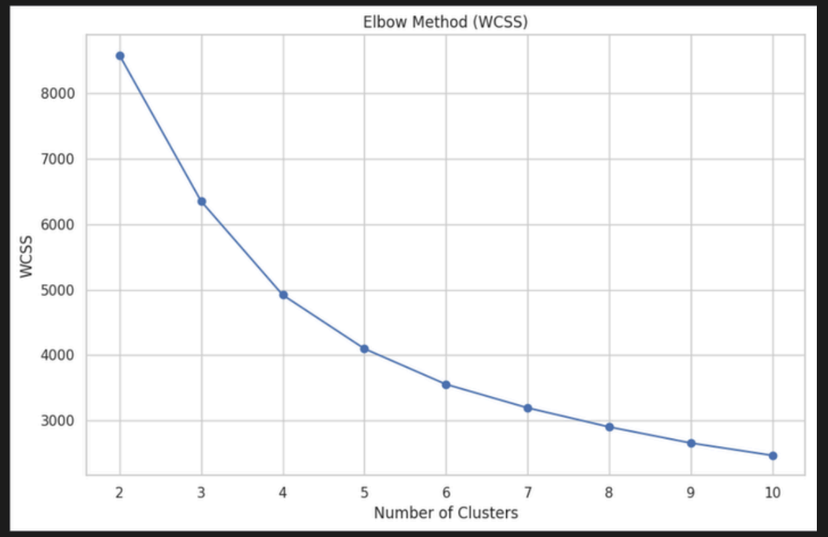
M — Monetary
Total amount spent

→ *paste RFM correlation heatmap here*

Customer Segmentation

APPROACH

- Log-transform + StandardScaler on RFM data
- Elbow method + silhouette score (~0.58)
- PCA for 2D cluster visualization



CUSTOMER SEGMENTS

HIGH-VALUE
Recent, frequent, high spend — loyalty targets

FREQUENT BUYERS
Repeat purchasers, moderate spend

AT-RISK
Lapsed — re-engagement targets

LOW ENGAGEMENT
Infrequent, possible one-time buyers

Predictive Modeling

MODEL WORKFLOW

RFM + behavioral features

Train / test split

XGBoost classifier

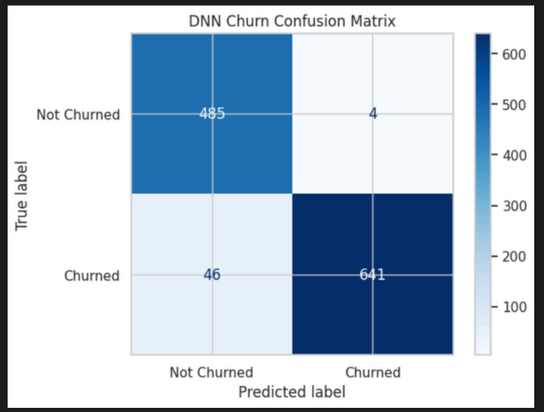
Accuracy · confusion matrix · report

WHY XGBOOST

- High accuracy on tabular retail data
- Handles large datasets efficiently
- Captures non-linear feature relationships

DNN COMPARISON

TensorFlow / Keras used for benchmarking and experimentation alongside XGBoost



DEMO

Next Steps

- Build an autoencoder to automatically flag customers with unusual purchase behavior
- Connect to live retail data for real-time predictions
- Integrate an LLM to generate dynamic, personalized offer messages

Thank You

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